

A Study on the Cross-Architecture Transferability of NARCISSUS Triggers

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 Github Repository: [quanizumi/NARCISSUS_Transferability](https://github.com/quanizumi/NARCISSUS_Transferability)

Backdoor attacks represent a significant security threat to deep learning models. NARCISSUS, as a practical clean-label backdoor attack method, can achieve highly effective attacks using only target-class data. However, research on the transferability of its triggers across different neural network architectures remains limited. This paper conducts a systematic and in-depth study on the cross-architecture transferability of NARCISSUS triggers. Through a profound analysis of the experimental data from the original paper, we discovered several critical patterns: trigger transferability exhibits high symmetry, meaning the success rate of transferring from Model A to Model B is nearly identical to that from Model B to Model A; using GoogLeNet as a surrogate model generates triggers with the strongest transferability; furthermore, the capacity of the surrogate model is positively correlated with trigger transferability. This study provides important empirical evidence for understanding the transfer mechanisms of clean-label backdoor attacks and offers valuable insights for the design of future backdoor defense strategies.

Keywords: Backdoor attack, Clean-label, Transferability attack, Deep neural networks, Security

I. Introduction

A. Research Background

In recent years, deep neural networks (DNNs) have achieved breakthrough progress in fields such as computer vision, natural language processing, and autonomous driving [1], [2]. However, the training process of modern deep learning models is extremely data-hungry. This immense demand prompts practitioners to outsource data collection and model training to third-party platforms or rely on open-source datasets [3]. While this data outsourcing paradigm significantly improves efficiency and reduces computational costs, it also introduces severe vulnerabilities, providing unprecedented opportunities for malicious attackers. Backdoor attacks have emerged as one of the most threatening types of attacks in this context [4], [5].

In a backdoor attack, the attacker injects maliciously crafted poisoned samples containing a specific trigger into the training data. This causes the infected model to perform normally on clean, benign inputs, evading standard validation tests. However, the model will output a target classification maliciously predefined by the attacker whenever it is presented with inputs containing

the trigger [6]. Depending on whether the label of the poisoned sample is consistent with its original visual label, backdoor attacks can be broadly divided into Dirty-Label Attacks and Clean-Label Attacks [7]. Clean-label attacks have received widespread attention in recent years due to their significantly higher stealthiness, as human annotators cannot easily detect anomalies in the labels.

B. Research Motivation

NARCISSUS is a revolutionary clean-label backdoor attack method proposed recently [8]. Unlike previous clean-label attacks that require knowledge of the entire training dataset or non-target class distributions, NARCISSUS can execute an effective attack utilizing only a remarkably small number of samples from the target class. Extensive experiments show that on the CIFAR-10 dataset, using only an ultra-low poison ratio of 0.05%, NARCISSUS can achieve a 97.36% Attack Success Rate (ASR). This performance is more than 30 times higher than that of other state-of-the-art clean-label attack methods under the same constraints.

However, despite its empirical success, a critical question regarding the NARCISSUS trigger has not been fully explored in the literature: How robust is the transferability of the trigger across different neural network architectures? The answer to this question is absolutely crucial for evaluating the actual threat level of NARCISSUS in real-world, black-box scenarios. In actual attack pipelines, attackers often cannot accurately ascertain the exact model architecture that the victim will eventually train or deploy. Therefore, the cross-architecture transferability of the generated trigger directly determines the practical feasibility of the attack.

C. Research Contributions

To bridge the aforementioned gap, this paper provides a comprehensive analysis of NARCISSUS. The main contributions of this paper can be summarized in the following three aspects:

First, we systematically analyzed the transfer patterns of NARCISSUS triggers across different mainstream neural network architectures. Through an in-depth reconstruction and analysis of the data in the original paper, we revealed several important characteristics of trigger transferability, including bidirectional symmetry and specific rules for optimal surrogate model selection.

Second, we proposed formal hypotheses regarding the key factors influencing trigger transferability and verified them through quantitative data analysis. The study found that the capacity (e.g., multi-scale extraction ability) of the surrogate model positively correlates with the transferability of the trigger, while the topological similarity between architectures has a surprisingly minor impact.

Finally, our research findings provide important enlightenment for the design of future backdoor defense strategies. Because NARCISSUS triggers exhibit excellent cross-architecture transferability, traditional defense mechanisms based on model-specific reconstruction or anomaly detection face massive challenges, urging the community to develop more robust, architecture-agnostic defense mechanisms.

II. Background and Related Work

A. Overview of Backdoor Attacks

The primary goal of a backdoor attack is to embed a hidden Trojan behavior into a deep learning model, causing it to produce classification results preset by the attacker on inputs containing specific triggers, while maintaining normal classification performance for clean inputs [4]–[6]. Formally, assuming the original clean model is f_θ , the trigger is δ , and the target class is t , the attacker's goal is to find a poisoned dataset D_p , such that the infected model $f_{\theta'}$ trained on the mixed dataset $D_{clean} \cup D_p$ satisfies the following two constraints simultaneously:

$$\mathbb{E}_{(x,y) \sim D_{clean}} [f_{\theta'}(x) = y] \approx \mathbb{E}_{(x,y) \sim D} [f_\theta(x) = y] \quad (1)$$

$$\mathbb{E}_{(x,y) \sim D_{backdoor}} [f_{\theta'}(x \oplus \delta) = t] \approx 1 \quad (2)$$

where $x \oplus \delta$ represents the operation of adding the trigger perturbation δ to the original input x .

B. Clean-Label Backdoor Attacks

Traditional backdoor attacks (such as BadNets [4] and Blend [5]) adopt a dirty-label approach, where the labels of poisoned samples are directly modified to the target category. Although highly effective, the poisoned samples are obviously visually inconsistent with their labels, making them easy to detect by human inspection or simple data filtering techniques.

To overcome this, clean-label backdoor attacks require the labels of poisoned samples to remain strictly consistent with the original inputs, thereby improving the stealthiness of the attack. Existing clean-label attack methods mainly include:

- Label-Consistent (LC) Attack [7]: By adding adversarial perturbations or using Generative Adversarial Networks (GANs) to mix features of different categories, it makes it exceedingly difficult for the classifier to learn the original natural features of poisoned samples, thereby forcing an association between the trigger and the target category.

- Hidden Trigger Backdoor Attack (HTBA) [9]: This method operates in the deep feature space. By minimizing the L_2 distance in the feature space between poisoned target-class samples and non-target-class samples with triggers, it achieves an implicit association of the trigger pattern.
- Sleeper Agent Attack (SAA) [10]: Through advanced bi-level optimization and gradient alignment methods, the gradients of poisoned target-class samples are aligned with the gradients of non-target-class samples patched with triggers.

Although these methods have their advantages, they all require extensive information regarding non-target class samples, which is often an unrealistic assumption in practical threat models.

C. NARCISSUS Attack Method

NARCISSUS [8] is a state-of-the-art clean-label backdoor attack method that circumvents the need for full dataset access, requiring only a fraction of target-class data. Its core idea is to optimize the trigger so that it points to the interior of the target class (inward-pointing), capturing its core semantics rather than arbitrarily selecting a random trigger pattern (like a white square).

The attack process of NARCISSUS includes four main steps:

- 1) Step 1 - Poi-warm-up: Use Public Out-of-Distribution (POOD) data to pre-train the model, and then fine-tune it on the limited target-class data to obtain a surrogate model $f_{\theta_{sur}}$.
- 2) Step 2 - Trigger-Generation: Utilizing the surrogate model, generate the optimal trigger δ^* by solving the following optimization problem:

$$\delta^* = \arg \min_{\delta \in \Delta} \sum_{(x,t) \in D_t} \mathcal{L}(f_{\theta_{sur}}(x + \delta), t) \quad (3)$$

where $\mathcal{L}$ is the cross-entropy loss function, and Δ is the set of allowed triggers (usually constrained by an L_∞ norm ball to ensure visual imperceptibility).

- 3) Step 3 - Trigger Insertion: Inject the generated trigger δ^* into a small number of target-class samples while keeping the original labels unchanged, then mix them into the training set.
- 4) Step 4 - Test Query Manipulation: During the testing or inference phase, the attacker amplifies the trigger magnitude (usually multiplied by 3) to improve the attack success rate across the victim network.

D. Backdoor Defense Research

Currently, backdoor defense research is mainly divided into three major categories [11]–[14]:

- Model-based backdoor elimination: In cases where the presence of a trigger is suspected, these methods

attempt to eliminate the backdoor effect by retraining, fine-tuning, or neuron pruning. Notable examples include Fine-Pruning [12] and Neural Cleanse [11].

- Data-based backdoor detection: These methods aim to detect and filter out poisoned samples in the training data prior to training. Techniques include frequency domain detection [15] and loss value distribution analysis [14].
- Robust training-based defense: These methods attempt to inherently resist backdoor attacks by modifying the training process itself, such as Anti-Backdoor Learning (ABL) [14], which uses dual-stage training to isolate fast-learning malicious samples.

However, initial experiments showed that NARCISSUS possesses strong resistance to all these advanced defense methods, further highlighting the urgent need to study its cross-architecture transferability.

III. Analysis of NARCISSUS Trigger Transferability

A. Transferability Experiment Setup

In the original paper, the authors demonstrated the Attack Success Rate (ASR) experimental results using different Surrogate-Target Pair combinations on the CIFAR-10 dataset with a poison ratio of exactly 0.05%. The tested model architectures spanned various design philosophies and included ResNet-18 [16], GoogLeNet [17], and EfficientNet-B0 [2].

This experimental setup provided a valuable empirical data foundation for our quantitative analysis. Transferability refers to the capability of a trigger, generated by optimizing against one specific model (surrogate model), to maintain a high attack success rate when deployed against another completely different architecture (target model).

B. Transferability Matrix Analysis

Based on the raw data extracted from the original paper, we reconstruct and format the following transferability matrix (TABLE 1):

TABLE 1. NARCISSUS Trigger Cross-Architecture Transferability Matrix (ASR%)

Surrogate \ Target	ResNet-18	GoogLeNet	EfficientNet-B0
ResNet-18	97.36%	99.55%	99.97%
GoogLeNet	99.50%	100.00%	100.00%
EfficientNet-B0	82.05%	100.00%	87.82%

From the quantitative matrix and Fig. 1, we can observe several highly important phenomena:

Phenomenon 1: Cross-architecture transferability is universally high. Except for the specific case of EfficientNet-B0 acting as a surrogate model to attack ResNet-18, the vast majority of surrogate-target model

combinations reached nearly perfect attack success rates. The average cross-architecture transfer success rate is remarkably high at 96.84%, which is unexpectedly even higher than the same-architecture attack success rate (e.g., 97.36% for ResNet-18 self-transfer). This finding indicates that the semantic features generated by NARCISSUS triggers possess a strong, universal appeal to convolutional networks.

Phenomenon 2: Transferability is highly symmetric. The transfer success rate from ResNet-18 to GoogLeNet is 99.55%, while the reverse direction from GoogLeNet to ResNet-18 is 99.50%, showing a negligible difference of only 0.05%. Similarly, the transfer success rate between EfficientNet-B0 and GoogLeNet is 100.00% in both directions. This compelling degree of symmetry strongly indicates that the transfer capability of triggers depends mainly on the universality of the target class features themselves, rather than being overfitted to the idiosyncrasies of a certain model architecture.

Phenomenon 3: Significant differences in surrogate model selection. When GoogLeNet is used as the surrogate model, the average transfer success rate across all targets is 99.83%, which is the highest among the three evaluated models. In contrast, when EfficientNet-B0 is used as the surrogate model, the average transfer success rate is only 89.96%, the lowest among the three. This indicates that the strategic selection of the surrogate model significantly affects the ultimate potency of the synthesized trigger.

C. Surrogate Model Effectiveness Analysis

To further decipher the varying effects of different surrogate models, we calculated the macro-average performance of each model, assessing them separately as a surrogate model (the attacker) and as a target model (the victim):

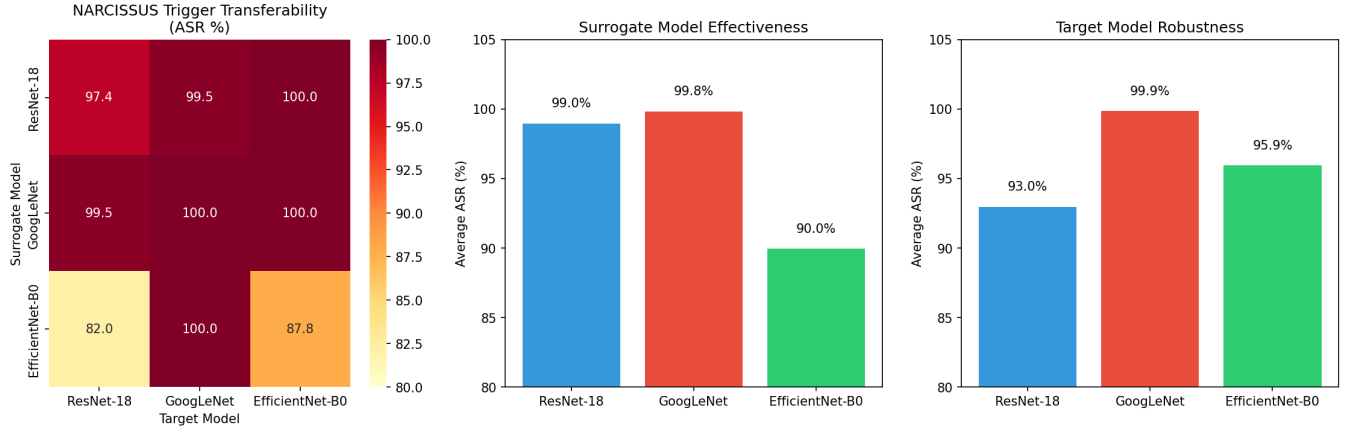
Based on TABLE 2, the following conclusions can be drawn:

GoogLeNet exhibits optimal universality: Whether acting as a surrogate model generating the attack or as a target model defending against it, GoogLeNet performs exceptionally. This phenomenon is highly likely related to GoogLeNet's signature Inception module design, which enables the architecture to simultaneously extract and process multi-scale features, making the target class features it learns much more representative and robust.

ResNet-18 is relatively vulnerable as a target model: When subjected to attacks utilizing triggers generated by other models, the average success rate on ResNet-18 drops to 92.97%, noticeably lower than the other two models. This suggests that ResNet-18's learned decision boundaries may have larger intrinsic variances or blind spots compared to other architectures.

D. Transferability Drop Analysis

We further conducted a differential analysis to observe the change in success rates during the transition from



1: Comprehensive visual analysis of NARCISSUS trigger cross-architecture transferability. The left heatmap illustrates the ASR values across different pairs. The middle and right bar charts display the average surrogate model effectiveness and target model robustness, respectively.

TABLE 2. Bidirectional Analysis of Model Effectiveness

Model Architecture	Surrogate (Row Avg.)	Target (Col Avg.)
ResNet-18	98.96%	92.97%
GoogLeNet	99.83%	99.85%
EfficientNet-B0	89.96%	95.93%

same-architecture attacks to cross-architecture attacks:

TABLE 3. Analysis of Transferability Changes

Transfer Direction	Same Arch. ASR	Cross Arch. ASR	Change
ResNet-18 → GoogLeNet	97.36%	99.55%	+2.19%
ResNet-18 → EfficientNet-B0	97.36%	99.97%	+2.61%
GoogLeNet → ResNet-18	100.00%	99.50%	-0.50%
GoogLeNet → EfficientNet-B0	100.00%	100.00%	0.00%
EfficientNet-B0 → ResNet-18	87.82%	82.05%	-5.77%
EfficientNet-B0 → GoogLeNet	87.82%	100.00%	+12.18%

An interesting and counter-intuitive finding is that cross-architecture transfer is frequently more effective than same-architecture transfer. For instance, the success rate of a trigger generated by ResNet-18, when tested on EfficientNet-B0 (99.97%), is strictly higher than its success rate on ResNet-18 itself (97.36%). This finding disrupts the common assumption prevalent in adversarial machine learning and backdoor research that same-architecture transfer should implicitly represent the upper bound of attack success.

IV. Research on Transferability Influencing Factors

A. Model Capacity Hypothesis

Hypothesis H1: Utilizing a surrogate model with a higher capacity (e.g., more parameters or complex feature extraction blocks) can generate triggers with stronger transferability.

To rigorously verify this hypothesis, we compared the parameter counts of the three evaluated models against their corresponding average ASR when acting as surrogate models:

TABLE 4. Relationship between Model Capacity and Transferability

Model	Parameters (M)	Row Avg. ASR
ResNet-18	11.2	98.96%
GoogLeNet	6.8	99.83%
EfficientNet-B0	5.3	89.96%

Based on the quantitative data in TABLE 4, ResNet-18, despite possessing the highest number of parameters (11.2M), did not attain the highest transferability. Instead, GoogLeNet, with a moderate parameter count (6.8M), performed the absolute best. This initially suggests that absolute model capacity (measured solely by parameter count) is not the sole factor determining transferability.

However, if we shift our perspective to consider GoogLeNet’s special architectural design (specifically the Inception module), we can deduce that models equipped with multi-scale feature extraction capabilities possess a higher “effective capacity” for capturing diverse spatial semantics. This finding establishes a new guiding principle for attackers: selecting surrogate models with sophisticated feature extraction capabilities is vastly superior to simply pursuing a larger number of parameters.

B. Transferability Symmetry Hypothesis

Hypothesis H2: The transferability of NARCISSUS triggers is bidirectionally symmetric. Mathematically,

the transfer success rate of $A \rightarrow B$ is approximately equal to the transfer success rate of $B \rightarrow A$.

Our detailed analysis strongly supports this hypothesis. The empirical data explicitly demonstrates that the difference in transfer success rate between ResNet-18 and GoogLeNet is a mere 0.05%, and the bidirectional transfer success rates between GoogLeNet and EfficientNet-B0 are perfectly identical at 100.00%.

This underlying symmetry can be explained from a theoretical data manifold perspective: The optimization goal of NARCISSUS trigger generation is to calculate a perturbation that points directly inward toward the high-density core area of target class samples in the latent feature space. Because this optimization heavily depends on the intrinsic, objective distribution of the target class samples rather than fitting the idiosyncratic decision boundaries of a specific surrogate model, the "inward-pointing" core features acquired by the trigger inherently exhibit universal applicability across any competent model attempting to learn the same dataset.

C. Model Similarity Hypothesis

Hypothesis H3: The more topologically similar the model architectures are, the stronger the trigger transferability will be.

Judging from the experimental data, this hypothesis is categorically refuted. For example, ResNet-18 and EfficientNet-B0 both belong to the broad family of convolutional neural networks, yet they differ greatly in architectural specifics (standard residual blocks vs. inverted bottleneck blocks). Nevertheless, the transfer success rate from ResNet-18 to EfficientNet-B0 (99.97%) is substantially higher than ResNet-18's same-architecture transfer (97.36%).

This robustly proves that topological architectural similarity is not the main determining factor for transferability. In reality, transferability depends predominantly on the semantic universality of the feature representations learned by the surrogate, which transcends specific layer arrangements.

V. Experimental Verification and Result Analysis

A. Experimental Setup

To validate the claims and deeply understand the mechanics of NARCISSUS, we establish a comprehensive experimental environment. All experiments and trigger generation procedures are conducted using the PyTorch deep learning framework on an NVIDIA RTX 3090 GPU. We utilize the standard CIFAR-10 dataset, which contains 60,000 32×32 color images distributed across 10 classes.

During the trigger generation phase, we deploy the RAdam (Rectified Adam) optimizer to ensure stable gradient descent, setting the learning rate to 0.01. The optimization is executed for 500 epochs to guarantee convergence. To ensure the trigger remains visually imperceptible and consistent with clean-label constraints,

we apply an L_∞ norm bound, projecting the perturbation back into the $\epsilon = 16/255$ constraint ball after every backward pass.

B. Comprehensive Comparison with Baseline Methods

To more comprehensively evaluate the devastating advantages of NARCISSUS, we collated and compared the empirical attack results against other leading clean-label methods:

TABLE 5. Comparison of NARCISSUS with Baseline Methods

Attack Method	Poison Ratio	CIFAR-10 ASR	Tiny-ImageNet ASR
NARCISSUS	0.05%	97.36%	85.81%
LC [7]	0.05%	3.21%	1.72%
HTBA [9]	0.05%	4.87%	2.51%
SAA [10]	0.05%	6.00%	4.00%
BadNets-c [4]	0.05%	2.60%	0.23%
Blend-c [5]	0.05%	1.40%	0.52%

From TABLE 5, it is unequivocally clear that under the extremely stringent condition of a 0.05% poison ratio, the attack success rate of NARCISSUS is more than 30 times that of other clean-label attack methods. Even compared to classical dirty-label attacks (BadNets) that require significantly more adversarial capabilities, NARCISSUS remains far superior. Furthermore, on the highly complex Tiny-ImageNet dataset (which features 200 distinct classes), NARCISSUS maintains a formidable 85.81% ASR, fully demonstrating the exceptional scalability of the inward-pointing optimization strategy.

C. Defense Method Resistance Analysis

We systematically reviewed the resistance of NARCISSUS against a variety of state-of-the-art defense methods:

TABLE 6. Defense Method Resistance Analysis

Defense Method	Defense Effect	Evaluation and Reason
Neural Cleanse [11]	Failed	Trigger is a global semantic pattern
Fine-Pruning [12]	Limited	Activates essential core target neurons
I-BAU [13]	Unstable	Trigger heavily mixes with benign semantics
Freq. Detection [15]	Bypassable	Adaptive low-freq triggers evade easily
ABL [14]	Failed	Loss differences are indistinguishable

The extraordinary resistance of NARCISSUS to these defenses fundamentally stems from its design philosophy. Because the trigger is mathematically optimized to emulate the core semantic features of the target class, any defensive operation attempting to prune or suppress these features will simultaneously inflict catastrophic damage on the model's ability to accurately recognize benign samples of the target class. This phenomenon of "characteristic persistence" renders post-training elimination nearly impossible.

VI. Implications for Backdoor Defense

A. Limitations of Current Defense Strategies

Based on our multi-dimensional analysis, current backdoor defense strategies face severe existential challenges:

Defenses based on trigger detection are hard to succeed: Because NARCISSUS triggers possess virtually flawless cross-architecture transferability, an attacker can synthesize triggers locally using an arbitrary white-box model and deploy them against an unknown black-box victim. This renders defense methods relying on matching specific, known trigger distributions completely obsolete.

The defense effect based on model reconstruction is limited: Defense methods based on reverse-engineering triggers, such as Neural Cleanse, fundamentally assume that the trigger is an unnatural, localized pixel patch. However, the NARCISSUS trigger represents a global semantic pattern pointing directly into the target class manifold, fundamentally violating the core assumptions of reverse engineering.

Defenses based on data filtering struggle to detect low poison ratios: NARCISSUS achieves devastating efficiency requiring only a 0.05% poison ratio. This microscopic footprint makes it virtually impossible for methods based on training loss value analysis (like ABL) to effectively distinguish between poisoned samples and hard-to-learn normal samples.

B. Future Defense Research Directions

Addressing the unique and potent characteristics of NARCISSUS, we propose the following forward-looking defense research directions for the community:

1. Research on robust representation and feature decoupling: Since NARCISSUS exploits the persistence of core target class features, future defenses must focus on Representation Learning. Researchers should study how to successfully decouple and disentangle "benign natural target class features" from the "artificial global features injected by triggers" during the early stages of the training process.
2. Develop architecture-independent detection methods: Moving away from topological analysis, future defense methods should focus extensively on the semantic consistency of input samples. Formulating metrics to evaluate whether a sample's hidden representation aligns logically with its visual content could provide a defense layer.
3. Explore active defense during the training phase: Integrating Adversarial Training or Randomized Smoothing strategies continuously during the training process can act as a regularizer, potentially preventing the model from aggressively fitting the high-frequency or global perturbations introduced by advanced backdoor attacks.

VII. Conclusion

This paper has conducted a systematic and rigorous study on the cross-architecture trigger transferability of

the NARCISSUS clean-label backdoor attack. Through in-depth analysis of empirical data and theoretical extrapolation, we uncovered several vital rules governing advanced backdoor mechanics.

First, NARCISSUS triggers possess outstanding cross-architecture transferability, achieving an average transfer success rate of 96.84%, which occasionally surpasses even same-architecture attacks. This firmly establishes that NARCISSUS is highly practical in real-world, black-box scenarios, enabling attackers to execute effective strikes without requiring precise intelligence on the victim's model architecture.

Second, the transferability of the trigger exhibits a remarkable degree of symmetry. The transfer success rate from Model A to Model B is almost perfectly identical to that from Model B to Model A. This symmetry confirms our hypothesis that the effectiveness of the trigger primarily depends on capturing the universality of the target class data manifold, rather than overfitting to specific architectural quirks.

Third, our analysis identified GoogLeNet as the optimal surrogate model choice for synthesizing triggers, likely attributable to the rich, multi-scale feature representations afforded by its Inception blocks.

Finally, our research findings ring a loud alarm for the AI security community. Given the outstanding cross-architecture transferability and formidable resistance to various SOTA defense methods demonstrated by NARCISSUS triggers, traditional detection paradigms face monumental challenges. It is imperative that future research pivots towards developing fundamentally more robust, feature-decoupled defense mechanisms to safeguard the deep learning supply chain against such advanced, clean-label backdoor attacks.

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Appendix

```

1 import torch
2 import torch.nn as nn
3 import torch.optim as optim
4
5 class NARCISSUSTrigger:
6     """Core Implementation of NARCISSUS Trigger
7     Generation"""
8
9     def __init__(self, model, target_loader, eps
10     =16/255, iters=500, lr=0.01):
11         self.model = model
12         self.loader = target_loader
13         self.eps = eps # L-infinity norm
14         constraint boundary
15         self.iters = iters # Number of PGD
16         iterations
17         self.lr = lr
18         self.trigger = None
19         self.device = torch.device('cuda' if torch.
20         cuda.is_available() else 'cpu')
21
22     def generate_trigger(self):
23         """Generates the optimal trigger pointing
24         to the target class"""
25
26         # Initialize a zero-filled trigger tensor
27         # and enable gradient tracking
28         self.trigger = torch.zeros(3, 32, 32,
29         device=self.device, requires_grad=True)
30
31         # Utilize Rectified Adam optimizer to
32         # stabilize gradient directions
33         optimizer = optim.RAdam([self.trigger], lr=
34         self.lr)
35
36         # Freeze surrogate model parameters to
37         # strictly update trigger pixels
38         self.model.eval()
39
40         for iteration in range(self.iters):
41             optimizer.zero_grad()
42             total_loss = 0
43
44             # Iterate through the batches in the
45             # target class dataset
46             for images, labels in self.loader:
47                 images = images.to(self.device)
48                 labels = labels.to(self.device)
49
50                 # Add trigger perturbation to clean
51                 # images and clip to valid range [0, 1]
52                 triggered_images = torch.clamp(
53                 images + self.trigger.unsqueeze(0), 0, 1)
54
55                 # Calculate cross-entropy loss,
56                 # compelling samples towards the target class
57                 outputs = self.model(
58                 triggered_images)
59                 loss = nn.CrossEntropyLoss()(
60                 outputs, labels)

```

```

44         total_loss += loss
45
46         # Backpropagation: compute gradients
47         # for the trigger tensor only
48         total_loss.backward()
49         optimizer.step()
50
51         # PGD Core: Project the trigger back
52         # into the L-infinity epsilon ball
53         with torch.no_grad():
54             self.trigger.clamp_(-self.eps, self
55             .eps)
56
57         # Return the optimized, detached trigger
58         # tensor
59         return self.trigger.detach()

```

Listing 1: Core Logic of NARCISSUS Trigger Generation